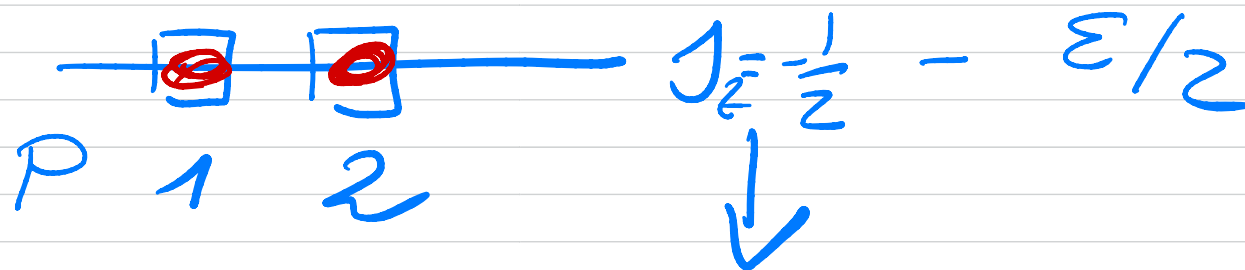
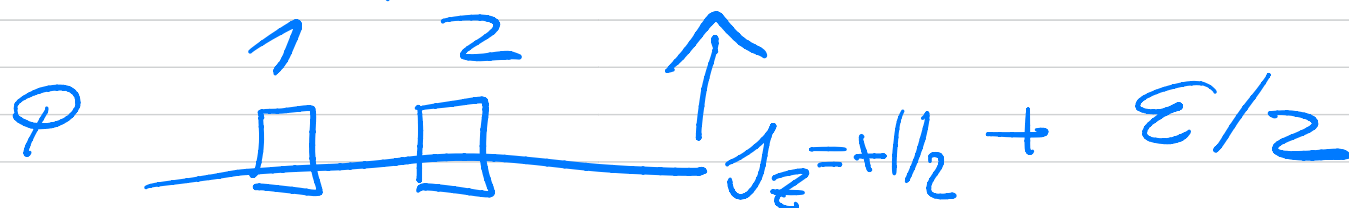


FYS5419/9419, lecture March 12

FYS5419/9419 March 12

Lipkin model

$N = 2$, two-level model



states

$$= \binom{4}{2}$$

$$= \frac{2 \cdot 3 \cdot 2}{1 \cdot 2 \cdot 1} = 6$$

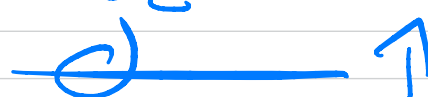
$$|4\rangle = |p\rangle \otimes |J_z\rangle = |pJ_z\rangle$$

Total spin projection

$$J_z = \sum_{i=1}^2 J_z^i$$

$$J_z = 0$$


A horizontal line representing an orbital with a circle in the middle. An upward arrow is to the right of the circle.

$$J_z = 0$$


A horizontal line representing an orbital with a circle in the middle. An upward arrow is to the right of the circle.

$$J_z = 0$$

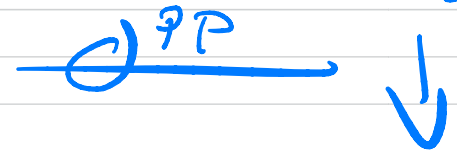

A horizontal line representing an orbital with a circle in the middle. An upward arrow is to the right of the circle. To the right of the circle, the text $|1, 1\rangle \oplus |1, 1/2 - 1/2\rangle$ is written.



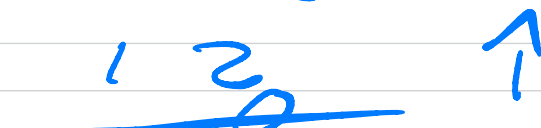
A horizontal line representing an orbital with a circle in the middle. A downward arrow is to the right of the circle.



A horizontal line representing an orbital with a circle in the middle. A downward arrow is to the right of the circle.



A horizontal line representing an orbital with a circle in the middle. A downward arrow is to the right of the circle. To the right of the circle, the text 3P is written.

$$J_z = 0$$


A horizontal line representing an orbital with a circle in the middle. An upward arrow is to the right of the circle. To the left of the circle, the numbers 1 and 2 are written vertically.



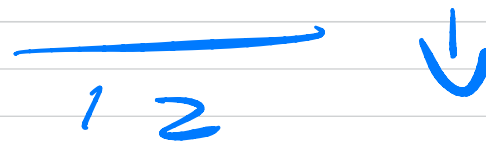
A horizontal line representing an orbital with a circle in the middle. An upward arrow is to the right of the circle. To the left of the circle, the numbers 1 and 2 are written vertically.

$$J_z = +1$$

P



A horizontal line representing an orbital with a circle in the middle. A downward arrow is to the right of the circle. To the left of the circle, the numbers 1 and 2 are written vertically.



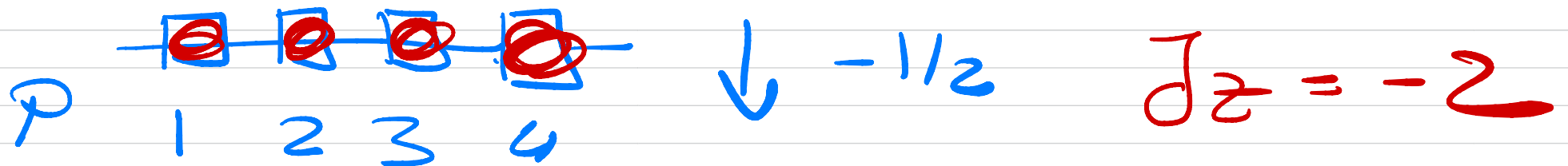
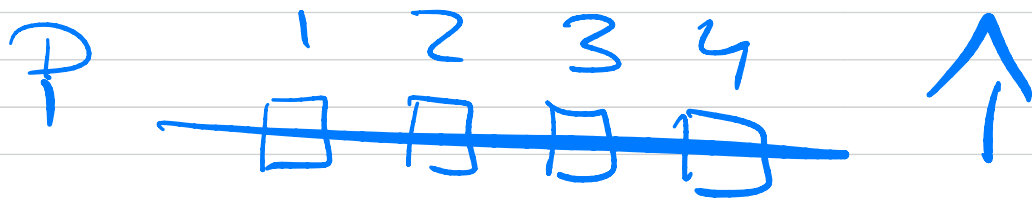
A horizontal line representing an orbital with a circle in the middle. A downward arrow is to the right of the circle. To the left of the circle, the numbers 1 and 2 are written vertically.

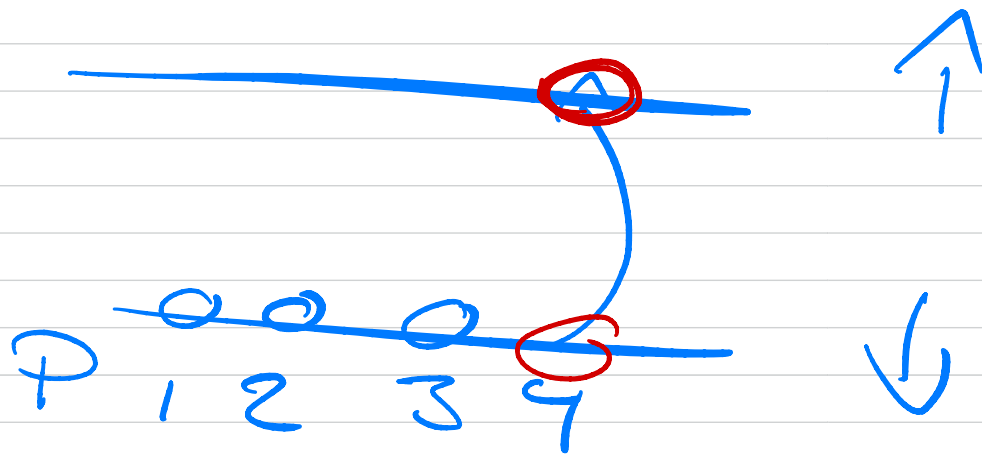
$$J_z = -1, 0, \underline{1} \quad \Rightarrow$$

$$J = \underline{1}$$

$$J_z = -J, -J+1, \dots, 0, 1, \dots, J-1, J$$

$$N = 4$$





$$J_z = -1$$

$$\# \text{ states} = \binom{8}{4}$$



$$J_z = 0$$



$$J_z = +2$$

...

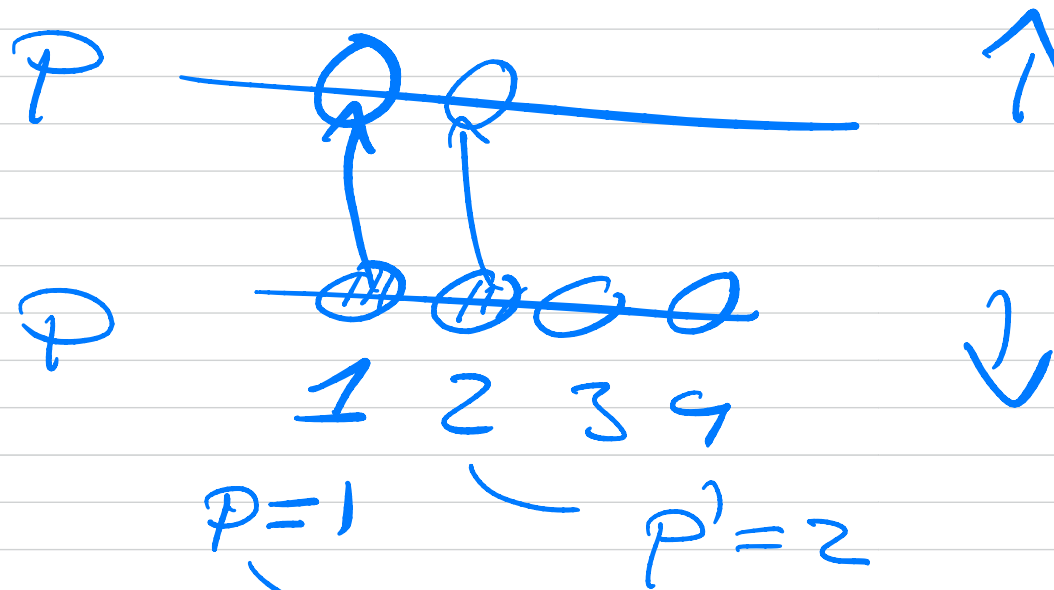
$$N = 4 \quad J_z = \{-2, -1, 0, +1\}$$

$$\Rightarrow J = 2 \quad \text{with degeneracy } 2J + 1$$

$$\mathcal{H} = \frac{\Sigma}{2} \sum_{\substack{J_z p \\ \{-1/2, 1/2\}}} J_z a_p^{\dagger} J_z a_p J_z$$

$$- \frac{V}{2} \sum_{J_z p p'} a_{p J_z}^{\dagger} a_{p' J_z}^{\dagger} a_{p' -J_z} a_{p -J_z}$$

$$- \frac{W}{2} \sum_{J_z p p'} a_{p J_z}^{\dagger} a_{p' -J_z}^{\dagger} a_{p' J_z} a_{p J_z}$$



$$-\frac{V}{2} \sum_{J_z P P'} c_{P J_z}^+ c_{P' J_z}^+ \times c_{P' - J_z} c_{P - J_z}$$

can rewrite in terms
of spin operators and
Pauli x, y, z and I
matrices

$$H_0 = \varepsilon J_z = \varepsilon \sum_{i=1}^N J_z^i$$

$$H_1 = -\frac{1}{2} V (J_+^2 + J_-^2)$$

$$J_{\pm} = J_x \pm i J_y$$

$$H_2 = -\frac{1}{2} U (J_+^2 - J_-^2 + N)$$

$$J_z = \frac{1}{2} \sum_{i=1}^N z_i \quad z_i = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$J_{\pm} = \frac{1}{\sqrt{2}} \sum_{i=1}^N (x_i \pm i y_i)$$

$$\begin{matrix} \swarrow \\ \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \end{matrix}$$

$$\begin{matrix} \downarrow \\ \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \end{matrix}$$

$$\mathcal{H} = \frac{\omega}{2} \sum_{i=1}^N z_i$$

$$= \frac{\omega}{2} \sum_{i < j} (x_i x_j - y_i y_j)$$

$$= \frac{\omega}{2} \sum_{i < j} (x_i x_j + y_i y_j)$$

$$\text{For } N=2 \quad (W=0)$$

$$\frac{E}{2} [\vec{z} \otimes \vec{1} + \vec{1} \otimes \vec{z}]$$

$$- \frac{V}{2} [\vec{x} \otimes \vec{x} - \vec{y} \otimes \vec{y}]$$

$$[J_z, J_{\pm}] = \pm J_{\pm}$$

$$[J_+, J_-] = 2J_z$$

$$[J^2, J_z] = 0, \quad [H, J^2] = 0$$

$$J^2 = J_x^2 + J_y^2 + J_z^2 \quad [H, J_z] = 0$$

$$\hat{J}_z |JJ_z\rangle = J_z |JJ_z\rangle$$

$$N = \sum_{i=1}^N a_i^\dagger a_i$$

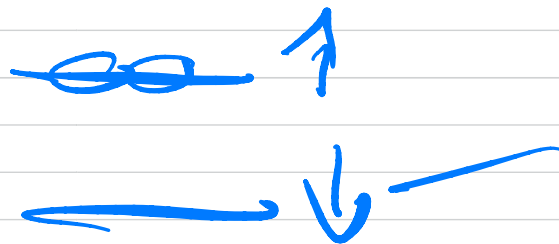
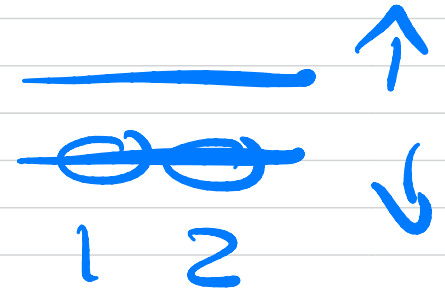
$$\hat{N} |JJ_z\rangle = N |JJ_z\rangle = 2J_z \times |JJ_z\rangle$$

$$\hat{J}_+ |JJ_z\rangle = \sqrt{J(J+1) - J_z(J_z+1)} \times |JJ_z+1\rangle$$

$$J_- |JJ_z\rangle = \sqrt{(J(J+1) - J_z(J_z-1))} \\ \times |JJ_z-1\rangle$$

$$N=2 \quad J=1 \quad J_z = \{-1, 0, 1\}$$

$$\langle \underset{J}{1} \pm 1 | \underset{J_z}{J_z} | 1 \pm 1 \rangle = \pm 1 |J=1, J_z=-1\rangle$$



$$|J=1, J_z=+1\rangle \\ |J=1, J_z=0\rangle$$

$$\langle 1 \pm 1 | \hat{J}_{\pm}^2 | 1 \mp 1 \rangle = 2$$

$$\langle 10 | \hat{J}_{+} \hat{J}_{-} | 10 \rangle = 2$$

$$\langle 10 | N | 10 \rangle = 2$$

$$\begin{bmatrix} -\epsilon & 0 & \sqrt{3} \\ 0 & \omega & 0 \\ \sqrt{3} & 0 & \epsilon \end{bmatrix}$$

	$ 1-1\rangle$	$ 10\rangle$	$ 1+1\rangle$
$\langle 1-1 $	$-\varepsilon$	0	ν
$\langle 10 $	0	w	0
$\langle 11 $	ν	0	z

$w = 0$

$$\begin{bmatrix} \nu & z \\ -\varepsilon & \nu \end{bmatrix}$$

~~$X \nu + z z -$~~

$$\begin{array}{c}
 \langle 1-1 | \\
 \langle 1 | \\
 \langle 10 |
 \end{array}
 \begin{array}{ccc}
 |1-1\rangle & |11\rangle & |10\rangle \\
 \hline
 \begin{array}{ccc}
 -\Sigma & V & 0 \\
 V & \Sigma & 0 \\
 0 & 0 & W
 \end{array}
 \end{array}$$

$$N=4 \quad J = 2 \wedge J_z \in \{-2, -1, 0, 1, 2\}$$

$$\langle 2 \pm 2 | \hat{J}_z | 2 \pm 2 \rangle = \pm 2$$

$$\langle 2 \pm 1 | \hat{J}_z | 2 \pm 1 \rangle = \pm \underline{1}$$

$$\langle 2 \pm 1 | \hat{J}_{\pm}^2 | 2 \mp 1 \rangle = 6$$

$$\langle 2 0 | \hat{J}_{\pm}^2 | 2 \mp 2 \rangle = 2\sqrt{6}$$

$$\langle 2 \pm 2 | \hat{J}_{\pm}^2 | 2 0 \rangle = 2\sqrt{6}$$

$$\langle 2 0 | \hat{J}_{\pm} \hat{J}_{\mp} | 2 0 \rangle = 6$$

$$\langle 2 \pm 1 | \hat{J}_{\pm} \hat{J}_{\mp} | 2 \pm 1 \rangle = 6$$

$$\langle 2 \pm 1 | \hat{J}_{\mp} \hat{J}_{\pm} | 2 \pm 1 \rangle = 4$$

$|2, 2\rangle$, $|2, 1\rangle$, $|2, 0\rangle$, $|2, -1\rangle$
 $|2, -2\rangle$

	$ 2, -2\rangle$	$ 2, -1\rangle$	$ 2, 0\rangle$	$ 2, +1\rangle$	$ 2, +2\rangle$
$\langle 2, -2 $	-2ε	0	$\sqrt{6}v$	0	0
$\langle 2, -1 $	0	$-\varepsilon + 3u$	0	$3v$	0
$\langle 2, 0 $	$\sqrt{6}v$	0	$4u$	0	$\sqrt{6}v$
$\langle 2, 1 $	0	$3v$	0	$\varepsilon + 3u$	0
$\langle 2, 2 $	0	0	$\sqrt{6}v$	0	2ε

J_{ℓ}''	$ 2-2\rangle$	$ 20\rangle$	$ 22\rangle$	$ 2-1\rangle$	$ 2+1\rangle$
$\langle 2-2 $	-2ε	$\sqrt{6}v$	0	0	0
$\langle 2\bar{0} $	$\sqrt{6}v$	$4w$	$\sqrt{6}v$	0	0
$\langle 22 $	0	$\sqrt{6}v$	2ε	0	0
$\langle 2-1 $	0	0	0	$-\varepsilon+3w$	$3v$
$\langle 2+1 $	0	0	0	$3v$	$\varepsilon+3w$

$$\begin{bmatrix} -\varepsilon+3w & 3v \\ 3v & \varepsilon+3w \end{bmatrix} = 3w \mathbf{I} - \varepsilon \mathbf{Z} + 3v \mathbf{X}$$

$$|2-2\rangle \rightarrow |00\rangle = |0\rangle \otimes |0\rangle$$

$$|20\rangle \Rightarrow |01\rangle \quad (110\rangle)$$

$$|22\rangle \Rightarrow |11\rangle$$

$$H'' = \begin{array}{c} \begin{array}{c} |00\rangle \quad |01\rangle \quad |10\rangle \quad |11\rangle \end{array} \\ \begin{bmatrix} -2\varepsilon & \sqrt{6}V & 0 & 0 \\ \sqrt{6}V & 4W & 0 & \sqrt{6}V \\ 0 & 0 & 0 & 0 \\ 0 & \sqrt{6}V & 0 & 2\varepsilon \end{bmatrix} \end{array}$$

$$\begin{bmatrix} 0 & \sqrt{6} & 0 & 0 & 0 \\ \sqrt{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$= \sqrt{6} \, V \, I, X_2 \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$X_2 = 1 \otimes X$$

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$= \sqrt{6} \vee \mathbb{I}_1, X_2 \underbrace{(\mathbb{I}_1 \mathbb{I}_2 + \mathbb{Z}_1 \mathbb{I}_2)}_2$$

$$\mathbb{Z}_1 = \mathbb{Z} \otimes \mathbb{1}$$

$$\mathbb{Z}_2 = \mathbb{I} \otimes \mathbb{Z}$$

$$X_1 = X \otimes \mathbb{1}$$

$$X_2 = \mathbb{1} \otimes X$$

$$\mathbb{I}_1 = \mathbb{I}_{2 \times 2} \otimes \mathbb{I}_{2 \times 2} = \begin{bmatrix} 1 & & 0 \\ & 1 & \\ 0 & & 1 \\ & & & 1 \end{bmatrix}$$

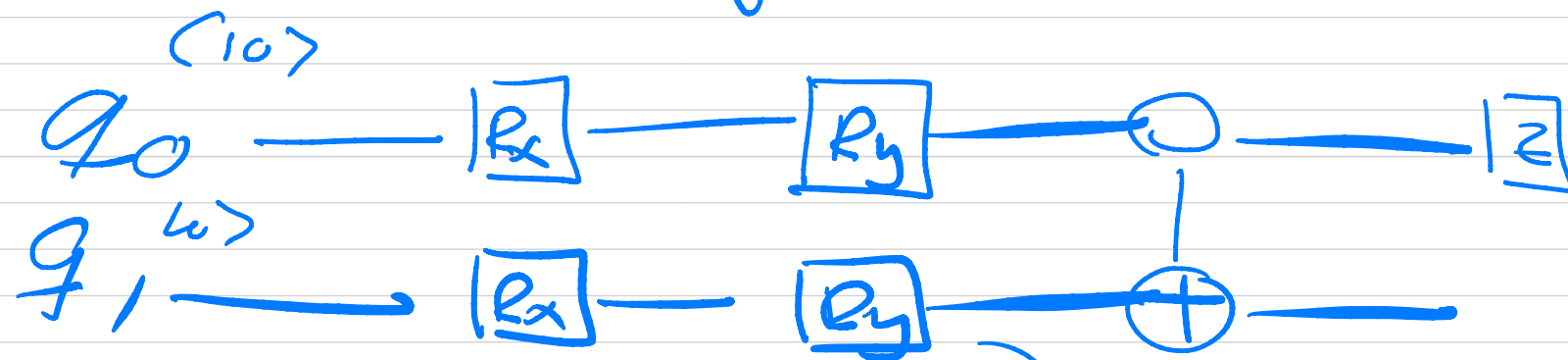
$$\begin{bmatrix} 0 & c & c & 0 \\ c & c & c & \sqrt{\delta} v \\ 0 & c & c & c \\ 0 & \sqrt{\delta} v & 0 & 0 \end{bmatrix} = \sqrt{\delta} v \sqrt{X \otimes Q\left(\frac{1-z}{2}\right)}$$

$$\begin{aligned} \mathcal{H} = & \quad w I_1 I_2 - w z_1 z_2 \\ & + (w - \varepsilon) z_1 - (w + \varepsilon) z_2 \\ & + \frac{\sqrt{\delta} v}{2} (x_2 + x_1 + z_1 x_2 - x_1 z_2) \end{aligned}$$

for project 4, feel free to
set $\omega = 0$ for $N=2$ and

$$N=4$$

ansatz for two qubit case



$$|q_0 q_1\rangle \propto$$

$$\propto (|00\rangle + |11\rangle)$$